#10 Traverse furthest branch, and update current values for NN and its distance.
#11 Return the closest point found so far.

As you can see, we check the distance of every intermediate point along the path, because any of them can be closer than the leaf: even if intermediate points cover larger regions than leaves, inside each region we don't have any indication of where dataset points might lie: if we refer to figure 9.20, if point `A` had been at `(0,0)`, the tree would have had the same shape, but `P` would have been closer to `A` (the root) than `G` (a leaf).

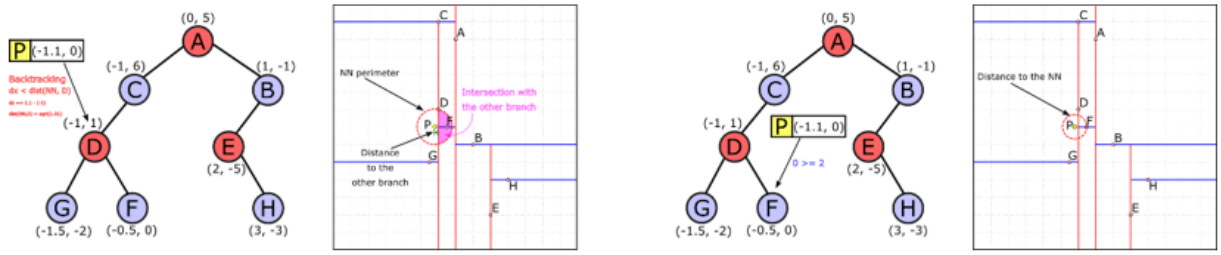


Figure 9.21 The second step, after an unsuccessful search, is to backtrack to check other branches of the tree for closer points. It is possible to have such points because when we traverse the tree, we cycle through the coordinates that we compare at each level, so we don't always go in the direction of the closest point, but we are forced to go on one side of the pivot, depending only on a single coordinate: so it is possible that the nearest neighbor is on the wrong side of the pivot with respect to our target point `P`. In this example, when we reached `D`, since it creates a vertical split, we needed to move to the left, as shown in figure 9.20: unfortunately the target point `P` and its